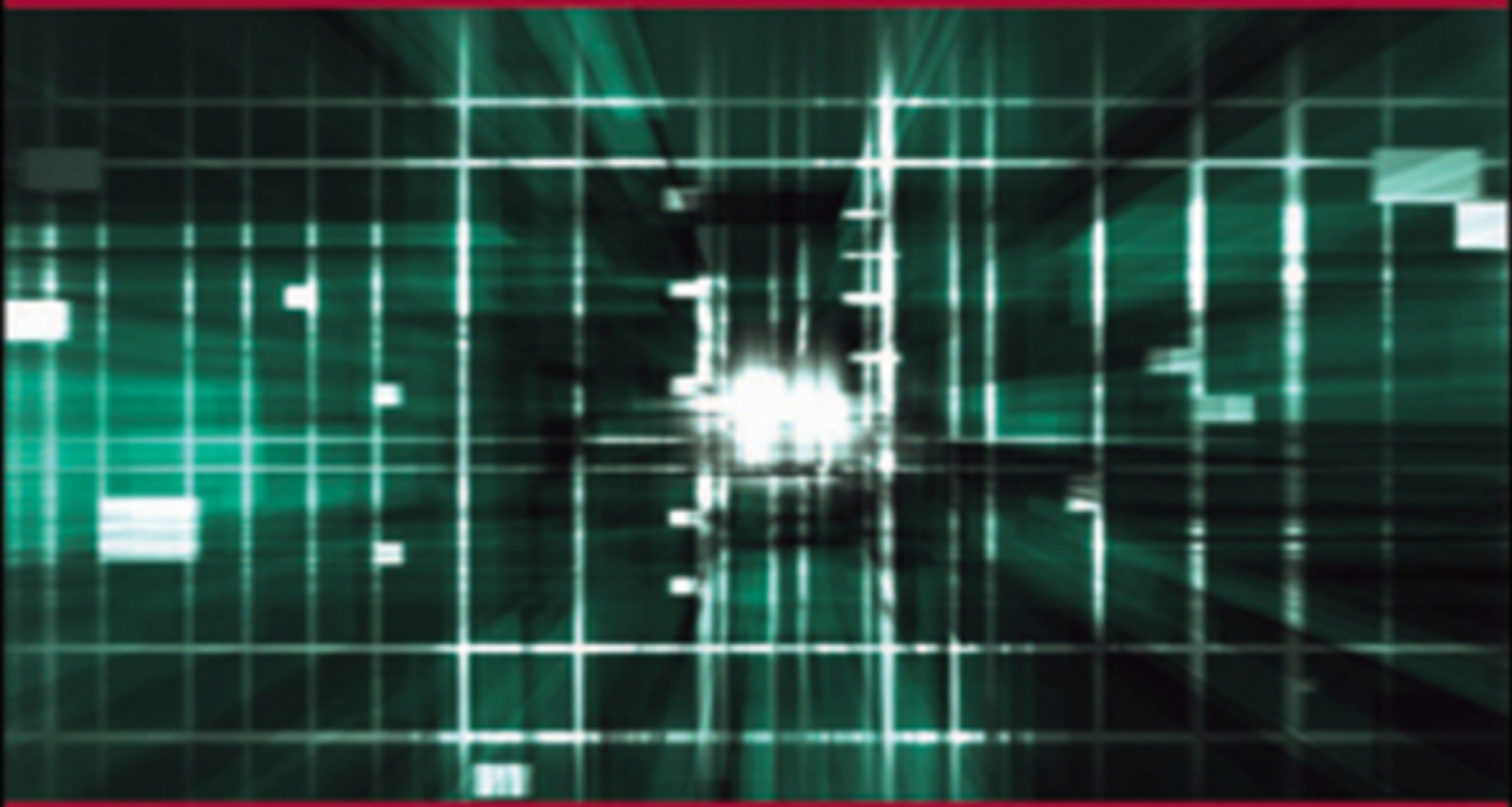
The Internet of Things

Connecting Objects

Edited by
Hakima Chaouchi



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Connecting Objects to the Web

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First published 2010 in Great Britain and the United States by ISTE Ltd and John Wiley & Sons, Inc.

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27-37 St George's Road
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John Wiley & Sons, Inc.
111 River Street
Hoboken, NJ 07030
USA

www.wiley.com

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Library of Congress Cataloging-in-Publication Data

The Internet of things : connecting objects to the web / edited by Hakima Chaouchi.
p. cm.
Includes bibliographical references and index.
ISBN 978-1-84821-140-7
1. Ubiquitous computing. 2. Computer networks. 3. Radio frequency identification systems. I. Chaouchi, Hakima.
QA76.5915.I67 2010
004--dc22

2010003706

British Library Cataloguing-in-Publication Data
A CIP record for this book is available from the British Library
ISBN 978-1-84821-140-7

Printed and bound in Great Britain by CPI Antony Rowe, Chippenham and Eastbourne.

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Paper from responsible sources
FSC® C013604

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Preface

Services designed over the Internet evolved depending on the needs identified from person-to-person interaction, such as email or phone services to meet other interactions, such as person-to-machine, machine-to-person and, lately, machine-to-machine where no human interaction is needed; thus building ubiquitous and pervasive computing. Such a computing system started a long time ago with the ambition of offering all-pervading computing to automate tasks and build a smart world. Introducing radio-frequency identification (RFID) technology in building new services over the network has pushed what is called the "Internet of Things" (IoT) as a meeting point between the real world and the virtual world, especially when combined with other technologies, such as sensor technology or mobile communication.

IoT appears to be one step further on the path to ubiquitous computing. This is possible with the introduction of RFID or sensor technologies, but also other technologies such as robotics, nanotechnology and others. These technologies make the Internet of things services an interdisciplinary field where most of the human senses are somehow reproduced and replaced in the virtual world.

So, what is meant by the Internet of things? From the economical point of view, it is about designing new services and generating new revenue streams in the communication value chain. This is not straightforward however, as lots of technical issues have been raised that need to be solved before an effective deployment of the new

envisioned services. From the technical point of view, it is about connecting new devices, called objects or things, and investigating the issues related to connecting these objects with the network in order to develop exploitable applications. To tackle these issues, it is important to understand what the Internet and things mean in the IoT, knowing that, depending on the research community, the meaning and the related issues might be different.

A thing or an object in the IoT is described as any item from our daily life that is enhanced with some computing and/or communication capabilities. For instance, items or objects with RFID or sensor technologies will become connected objects. These objects, depending on the application, might range from a size as small as an atom or as large as a building; they might be fixed or mobile, such as a car; they might be inanimate or animate, such as animals or humans. These objects joining an IoT service will have an electronic identification, such as an RFID. Objects or things are also new electronic devices interacting with the real-world environment, such as sensors.

Conventional communicating devices, such as laptops, computers and phones might be considered to be objects. In our book, we exclude these classical devices from the object list since they do not directly enable interaction with the real-world environment. Other objects, such as consumer electronic products like a TV or a fridge have already been introduced in the communication chain via other technologies, such as power line communication technology. IoT will clearly have to allow the connectivity of a large number and different types of objects. This means that it has to face the heterogenity and scalability of the communication framework in order to build the envisioned applications. These applications will orchestrate the real environment-related new functionalities of identifying, locating, sensing and acting, thus building the task automation and environmental task monitoring expected by the IoT.

Currently RFID technology and sensor technology are promising, very close-to-market applications, since they offer the new functionalities of identifying and sensing, respectively. Sensor

technology and sensor networks for phenomenon monitoring have interested the telecommunication research community earlier than RFID technology, which evolved in the retail product chain for product tracking and only recently joined the telecommunication value chain. Some recent examples show the development of RFID-based systems to help vision-impaired people to be guided on buses and enhance museum visits with smart phones and RFID. Combining RFID, sensor, and mobile communications appears to be very promising and will enable more applications to contribute in building the IoT. Although already used, these technologies need to be improved from security, privacy, performance and scalability points of view.

On the other hand, the Internet in the IoT might also have different interpretations. The obvious interpretation, which is more direct, refers to the current Internet adapted to these new objects' connectivity needs. Current Internet is that of connected nodes using a TCP/IP (the internet protocol suite) protocol stack with IP addressing and routing capabilities. Usually, the Internet model runs a TCP/IP stack in the connected device or offers the possibility of designing corresponding gateways to specific nodes or networks.

Connecting objects to the current Internet involves adapting the TCP/IP stack to the resources of the objects. This is what is proposed by the Internet Engineering Task Force with the 6LoWPLAN protocol stack for sensor networks. It also means designing gateways connecting the objects to the Internet, as might be done with connecting RFID objects to the Internet via gateways.

Another view of the IoT involves designing a new communication model, different from TCP/IP. This would be a new Internet, also called future Internet, where it is possible to adapt the communication model to the context, traffic constraints, resource limitations and so on. Note that designing the network of the future (or future Internet) is one of the major research goals of the current networking research community, where better network adaptation than the current Internet is expected.

In the long run, the IoT appears to be one of the leading paths to this goal since it challenges the current Internet model with new needs of object connectivity: such as identification, naming and addressing, scalability, heterogenity, resource limitation, new traffic modeling, etc.

While waiting for the future Internet, the current Internet operators show a great interest in the concretization of unlimited IoT services. They are welcoming any new and attractive internet services generating new traffic to be transported by the Internet or all-IP network that already offers one network model for multiservice support, such as voice, data and multimedia services.

Designing services involving real-world things' and objects' interaction and communication through the Internet is therefore highly encouraged under the condition of solving all the related issues of security and privacy and of connecting billions of objects to the Internet directly or through gateways. These gateways can be simple or intelligent gateways, capable of interpreting the traffic needs at the entrance of the network.

The current objects' resources such as memory, processing and battery in tiny objects are very limited and cannot run the current Internet communication model which means that these objects will use an adapted version of the Internet model, a proprietary communication system that will be seen as a heterogenous one from the Internet, and thus needs a gateway to benefit from the forwarding of traffic.

Of course, the traffic that will be generated by these object-based applications will have different expectations from the network. In fact, until now voice was considered the most difficult object to transport since it was used over a circuit-switched forwarding system designed to match its expectation. Now the traffic generated by these specific IoT applications will have to be modeled and need to be satisfied, probably partially by the current Internet and totally by the future Internet. For instance, if an "*actuate*" is ordered remotely via the network, the traffic priority should match the emergency of this

action. Also the packet size should be adapted to this new type of information. Similarly to the voice application, with the IoT-generated traffic, the packet design and priority of the packets will have to be specified to match the traffic requirement.

It is clear that plenty of technical, research, economic and societal issues are correlated with the IoT. In this book *The Internet of Things*, we have tried to bring together the up-to-date knowledge associated with what a connected object means, what Internet means in the IoT, and what the technical challenges (see Chapter 1) are with a more network-related view.

The book, also describes what the enabling technologies of the IoT are; the closest to the market are described in detail. These are mainly RFID (Chapter 2) for identifying and tracking the objects, and sensors (Chapter 3) for sensing the environment and actuating. Both RFID and sensor technologies use wireless connectivity.

This book additionally describes power line communication technology (Chapter 4) used for home networking. This applies the idea of building smart homes by connecting smart objects at home, such as a fridge and TV. This idea emerged before we started to use the IoT terminology, which was pushed more with RFID-connecting objects. Services developed in home networking are also part of the IoT services, but do not have the same connectivity issues as RFID or sensors, which are tiny devices with limited resources, mainly battery power.

This book, discusses the applications and research issues related to RFID (Chapter 5). It also proposes to look at other RFID technology usage in improving some network-related functionalities, such as location and mobility (Chapter 6). Finally, setting the standards and the governance of the IoT is discussed in Chapters 7 and 8.

We are not ignoring other issues related to the IoT, such as the need for high-performance computing to face scalability, the need for even faster processing and the limits of component physics in increasing the speed of processors, to face the expected billion

connected objects generating traffic in the network. Moreover, research disciplines will have to work and interact with the networking community to build ubiquitous computing and design the IoT services and networking.

Hakima CHAOUCHI

April 2010

Chapter 1

Introduction to the Internet of Things

1.1. Introduction

The Internet of Things (IoT) is somehow a leading path to a smart world with ubiquitous computing and networking. It aims to make different tasks easier for users and provide other tasks, such as easy monitoring of different phenomena surrounding us. With ubiquitous computing, computing will be embedded everywhere and programmed to act automatically with no manual triggering; it will be omnipresent.

In the IoT, environmental and daily life items, also named "things", "objects", or "machines" are enhanced with computing and communication technology and join the communication framework. In this framework, wireless and wired technologies already provide the communication capabilities and interactions, meeting a variety of services based on person-to-person, person-to-machine, machine-to-person, machine-to-machine interactions and so on. These connected machines or objects/things will be new Internet or network users and will generate data traffic in the current or emerging Internet.

Chapter written by Hakima CHAOUCHI.

Connecting objects might be wireless, as with the radio frequency identification (RFID), or sensor radio technologies that offers, respectively, identification of items and sensing of the environment. Connection may be wired, as with power line communication (PLC). PLC offers data transport over electrical media and has pioneered the in-home networking connectivity of electronic consumer devices that we also name "objects" such as smart fridges, smart TVs, smart heaters, etc.

IoT-based services will provide more automation of various tasks around people and connected objects in order to build a smart world not only in manufacturing industries but also in the office, at home and everywhere. Most of these services will also rely on the easy location and tracking of connected objects. Other services – object-to-object-oriented services – will emerge for instance in the context of the green planet goal. This is where specific applications will monitor the environment and automatically react, for example, to minimize energy wastage or avoid natural disasters.

In the IoT, identifying, sensing and automatically deciding and actuating will be the main new functionalities that will enable ubiquitous computing and networking. Therefore, sensor and RFID, among other technologies, will be increasingly deployed and will thus allow integration of the real world environment in the networked services. In fact, billions of RFID tags and sensors are expected to connect billions of items/objects/things to the network in the coming years. Scalable identification, naming and addressing space and structure, scalable name resolution, scalable and secure data transfer are all of major concern. Other enabling technologies for this real-world networked service include nanotechnology, automatic processing and robotics, and probably newly-emerging technologies enabling the envisioned smart world to become real.

IoT will connect heterogenous devices and will be very dense, connecting billions of objects. An Internet-, IP- (Internet protocol) or TCP/IP (transport control protocol/Internet protocol) -based model stands at the centre of the IoT. It is one possible INTERNETworking solution to hide the ever-increasing heterogenity of networking technologies and communication systems in the ubiquitous

environment envisioned. IP might not, however, support the resource limitation and scalability of the network.

IP or the Internet will certainly support the close-to-market IoT applications, but IoT research development will hopefully also come with a new INTERNETworking communication model and architecture. These will better support the new requirement of the heterogenity of objects, scalability (of billions of objects expected), limited resources of connecting objects and requirements related to new services and applications to be designed over this connected real world. It falls exactly under the post-IP or future Internet era [EUR 08, GEN 10, FIN 10], where several research projects are building a new communication model and architecture that is more adaptive to the requirements of a given network.

IoT is one network with new requirements related to the introduction of these nodes/objects with new technologies in the network. The existing TCP/IP model might be compatible with the emerging post-IP or future Internet model. While seeking the design of the IoT network and services, a rethinking of the basic concepts will emerge related to addressing, routing, scaling, guaranteeing quality of service, security, mobility, etc. These research projects are currently supported by the all-IP network, where the packet-switching TCP/IP model has taken over the classical telecom circuit-switching model. Thanks to convergent efforts, the Internet is already the generalized model in telecommunications to offer different services.

1.2. History of IoT

IoT was originally introduced by the Auto-ID research center at the MIT (Massachusetts Institute) [AUT] where an important effort was made to uniquely identify products. The result was termed EPC (electronic product code), which was then commercialized by EPCglobal. EPCglobal was created to follow the AutoID objectives in the industry, with the EAN.UCC (European Article Numbering – Uniform Code Council), now called GS1, as a partner to commercialize Auto-ID research, mainly the EPC.

A “thing” or “object” is any possible item in the real world that might join the communication chain. As presented by [HOD 01], the initial main objective of the IoT was to combine the communication capabilities characterized by data transmission. This was viewed as the Internet, also known as the network of bits representing the “digital world”. The process of automation was viewed as connecting the real or physical world, named the “network of atoms” characterized by the smallest component, which is the atom, to the digital world, named the “network of bits”, characterized by the smallest component, which is the bit.

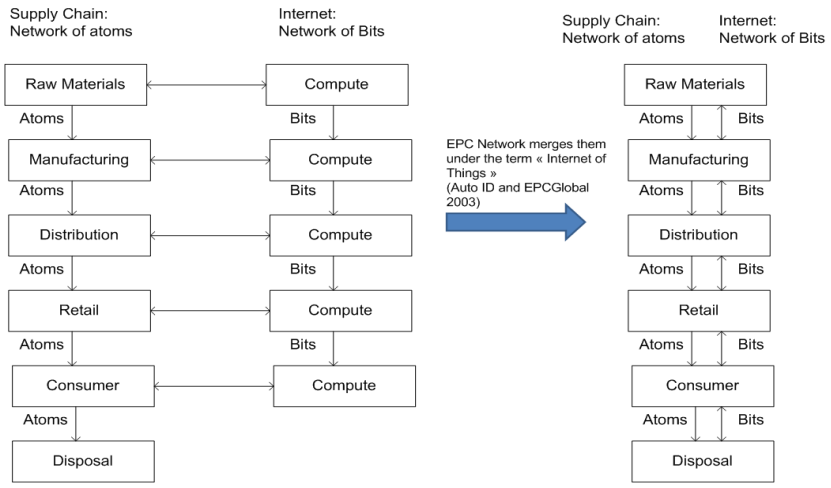

Figure 1.1. *Origin of IoT [HOD 01]*

In 2005, the ITU (International Telecommunication Unit) showed interest in new telecommunication business possibilities that could be built into services around the new connectivity of environment objects to the network.

The ITU produced a comprehensive report on the IoT from technical, economical and ethical views [IoT 05]. It introduced a new axis in the ubiquitous networking path to complete the existing “anywhere” and “anytime” connectivity. It is the “anything” connectivity axes where the thing-to-thing or machine-to-machine interaction is added to complete the existing person-to-person and

person-to-machine interaction in the possible connectivity framework. This clearly opens new service opportunities.

Figure 1.2 presents the ITU view of ubiquitous networking, adding the "anything connection" to the connectivity anywhere and anytime.

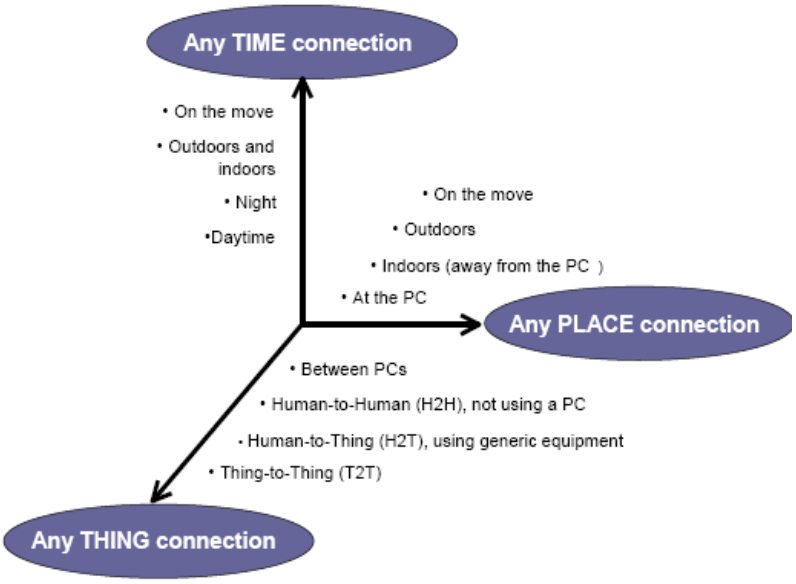

Figure 1.2. *ITU any place, any time and any thing vision [IoT 05]*

By adding the "any thing" connection axis, new sources of information are introduced in the connected network and this enables new services exploiting the newly-introduced information in the network. These services will be designed to offer the expected ubiquitous networking, where the real-world environment might react and adapt to different situations in order to make human life easier and more comfortable. Connecting these new objects will obviously raise many questions such as:

– the connecting technology of the so-called objects;

– the interoperability between objects;

– the communication model of these connected objects;

– the possible interaction with the existing models, such as the Internet;

– the choice of the transport model;

– the addressing, identifying and naming;

– the security and privacy;

– the economic impact and the telecommunication value chain evolution.

In fact, most of the Internet services were designed to satisfy person-to-person interaction, such as email and phone service. The traffic transported through the Internet is currently generated by people; either voice or data. New services were then developed around person-to-machine and machine-to-person interactions, such as video-on-demand or content distribution services. Finally, in order to provide tasks and process automation, new services will be developed around the machine-to-person, machine-to-machine or thing-to-thing and any other possible interactions in the so-called ubiquitous networking, as shown in Figure 1.3.

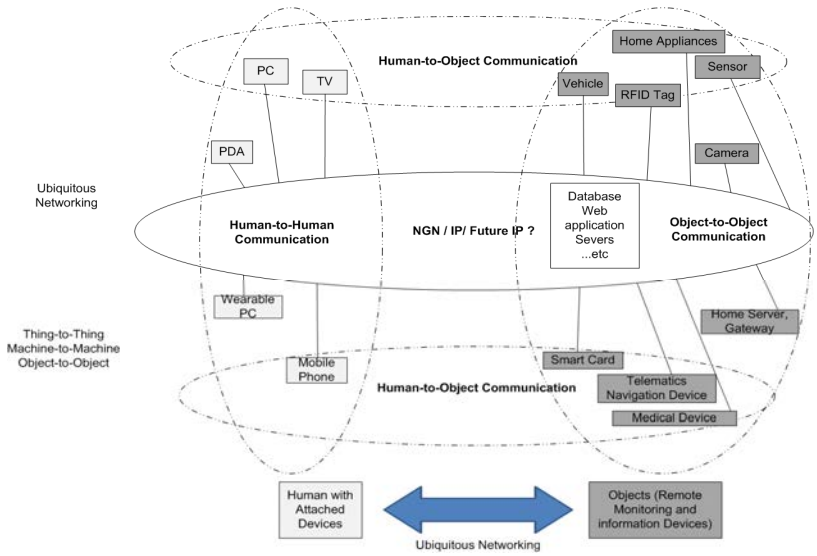

Figure 1.3. *Ubiquitous networking [IoT 05]*

IoT will connect objects to offer new services around people and objects; we can also call it the "Network of Things/Objects". IoT might suggest that the Internet model will have to be adapted to support the connectivity and traffic transport of new services based upon the connected objects. It is also worth mentioning that "Web of Objects" is another term used to refer to the IoT. As the Web is the main service accessibility to current Internet-connected nodes, similarly IoT is seen as the main service accessibility to the networked and connected objects. Also, in IoT, the naming resolution of identifiers to Web addresses is needed to handle the correspondence of identifiers introduced by RFID technology and ONS (object name service) has been introduced for that – as a similar service to the internet DNS (domain name service). "Web of Objects" has more meaning from the application viewpoint, without indirectly implying the extension of the Internet communication model to these new connected objects, as "IoT" might suggest.

1.3. About objects/things in the IoT

What exactly is a connecting or connected object or a thing? In close-to-market IoT applications, RFID tags and sensors are connecting inanimate objects and are building the actual things enabling the first IoT services.

Following the American Auto ID research center description of the IoT and the European CASAGRAS research project terminology [CAS 08], "things" or "objects" are described as a set of atoms. The atom is the smallest object in the IoT; as could be seen by nanotechnology, which is one of the enabling technologies of the IoT. A network of atoms combined with a network of bits falls into what is named the IoT. It will gather a set of objects connected to the network to help in the execution of new services enabling the smart world. So with the atom, being the smallest possible object, it is possible to classify objects based on their size and complexity, their moveable aspect and whether they are animate or inanimate, as shown in Figure 1.4 [CAS 08].

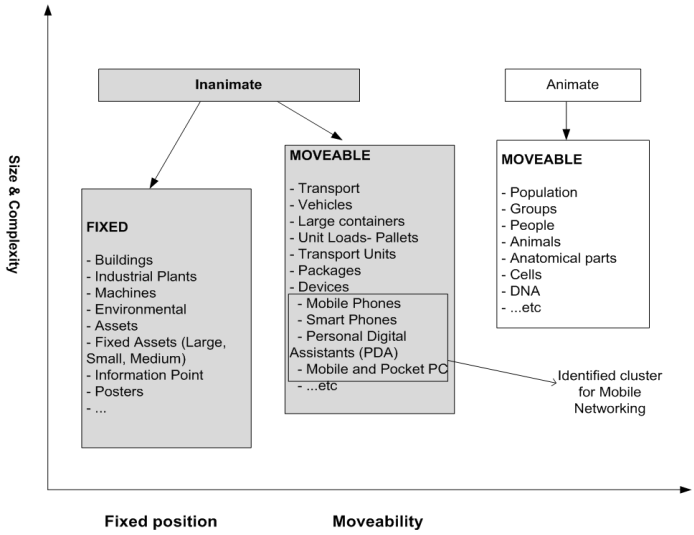

Figure 1.4. *Objects classification [CAS 08]*

In this terminology, classic devices such as PCs and mobile phones are already connected objects using wired or wireless communication. IoT will extend the connectivity and interworking of these currently existing objects with new objects connected through radio sensing or identifying technologies, such as sensor or RFID networks, allowing the development of new services involving information from the environment. This information could be either a simple identifier, as with RFID, or captured information, as with sensors. In other terminologies, common networking devices such as PCs, laptops and mobile phones are not considered to be objects.

Only small devices, such as sensors, actuators and RFID added to objects are considered as connected things or objects. Also, machines identified in home networking (connected consumer electronic devices, such as smart TVs, fridges, lights, etc.) are also connected objects. In this book, by “thing” or “object” we refer to daily life and surrounding items connected using radio connectivity, such as sensors, RFIDs or wired communication such as PLC. These technologies are enabling the development of new services, orchestrating real-world information via the connected objects.

Different technologies can be used to interconnect objects. Note that connecting objects, such as consumer electronics, e.g. a smart fridge or a smart heater, has started with home networking where consumer appliances are connected through wired technology, such as PLC, allowing communication through the power line. A number of standardization and industry organizations are addressing different issues of the home networking puzzle.

Current home networking applications do not suffer from any resource limitations. The connected objects (smart fridge, smart TV, etc.) can easily deploy an existing communication model, such as the TCP/IP model, to allow data transmission. They are affected more by interoperability problems. This is different from the issues of new applications of IoT, which rely on sensors and RFIDs where the resources of the connected objects via radio are limited by energy, memory and processing capability.

Another concern is how to support the connectivity of heterogenous objects, when a huge number of these objects/things will be connected by tags or sensors. Sensor networks have been used in industrial process control. They have allowed automation of the sense and actuate processes in order to perform automatic control, maintenance and data collection operations. A large number of potential environment monitoring applications for RFID and sensor networks are still to come. In home networking, new applications using sensor and RFID technologies will allow the automatic control of certain processes, hence minimizing human intervention.

1.4. The identifier in the IoT

IP addresses identify nodes in the Internet and serve as locators for routing. IPv6 allows larger address space than IPv4. In the IoT a large identification space will be needed to cover the identification of the tremendous number of connected objects. A specific semantic of these identifiers will follow the application's need. In the IoT, where objects are addressed via identifiers stored into tags and interrogated by networked readers, the question of unifying and standardizing the identifier's size and structure is critical in order to allow large

deployment of services relying on these new connected objects. Since RFID technology is naturally used for identification, the standardization of the identifier stored in the RFID is the current IoT concern. The same question is raised for any addressing schemes used in the network of objects. In the IP based case, the problem will be more about the semantics of the identifier, scalability of the addressing space and memory size limitation of the devices addressed by the chosen address/identifier space.

The term "identifier" is similar to the term "name". A name does not change with location, in contrast to an "address", which is intended to be used to refer to the location of a thing. IP addresses are used to route packets between end-systems. Emerging IoT service providers expect to rely on a convenient identifier space for the envisioned service, knowing that anything can be assigned an identifier – a physical object, person, place or logical object. A wide variety of services and applications can be envisaged once it becomes possible to provide information associated with a tag identifier in different forms (text, audio or image). For example, in a museum, an identifier on a tag attached to a painting could be used to find further information on the painting and the artist. In a grocery store, an identifier on a food package could be used to check that the food is safe to eat and not a member of a sample that has been found to be contaminated in some way. Other areas in which identifier-triggered information access could be valuable are in:

– medicine/pharmaceuticals;

– agriculture;

– libraries;

– the retail trade;

– the tourist industry;

– logistics; and

– supply chain management [MAI 10].

So, the major issue to start with to maximize success is the standardization in order to ensure interoperability of the connected

objects and nodes in the IoT. As will be presented in Chapter 7 of this book, this problem is well known in the communication field, but it is worse in the IoT as billions of objects are expected to be connected. It is therefore important to standardize the object identifier since the objects in the network will be addressed by a unique identifier similar to IP the addresses of connected nodes in the Internet.

EPCglobal first standardized the EPC identifier, followed by the International Standardization Organization (ISO). In addition to ISO and EPCglobal, the ubiquitous ID Center (uIDcenter) has defined a generic identifier called "ucode", which is not only intended to identify physical objects but also extended to places and digital information. ISO has addressed the issue of standardized identifiers by considering proprietary proposals, such as EPCglobal and uIDcenter, but it also offers the chance to define other identifiers that conform to ISO recommendations.

For example, if we use IP address space for identification, and if a device/thing has enough memory, we can consider IPv6 address space to be used as an identifier space of objects, since IPv6 address space is supposed to be large enough to offer up to 223 addresses in a square meter. Unfortunately, defining an identifier is not only about the scalability of the identifier space but is also about the structure and meaning/semantic of the identifier. It is important that an identifier only plays the role of identification, so that even if the objects identified are mobile, the identifier remains the same. In the IP communication model, IP addresses play two roles: from a network point of view, they act as a locator for routing and from an application point of view they identify hosts for the duration of a communication session. This dual role is seen to be problematic due to increasing demands for mobility and the multi-homing of end-systems.

For this reason the Internet Research Task Force (IRTF) and the Internet Engineering Task Force (IETF) have developed the host identity protocol (HIP), which defines host identifiers that can perform the identifier role of the IP address. This leaves the IP address to act solely as a locator for routing. These host identifiers of HIP protocol could potentially be used as another type of identifier in the IoT under the condition that they respect the ISO standard and are capable of